

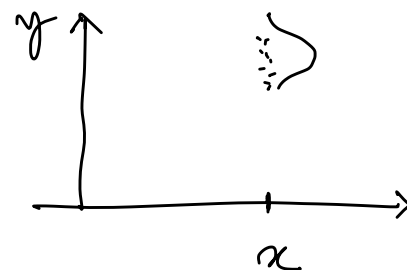
18/09/25

Bias - Variance Trade Off

Training Data $\mathcal{D} = \{ (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n) \}$ Regression Problem
 for a specific x_i , there can be multiple y_i 's $|D|=n$ $y_i \in \mathbb{R}$
 $x_i \Rightarrow$ Vector

Expected label (given $x \in \mathbb{R}^d$)

$$\bar{y}(x) = E_{y|x} [y] = \int y P(y|x) dy$$



Given $D \rightarrow$

$$\underline{h_D} = A(D)$$

A = Algorithm of the model

$$D \sim P(x, y)$$

$$D \sim P(y|x) P(x)$$

In the test time

Expected Test Error (Given h_D)

$$E_{(x,y) \sim P} [(h_D(x) - y)^2] = \int \int (h_D(x) - y)^2 P(x, y) dy dx$$

So far we assumed that there is only one D : training set
 D itself is sampled from P^n and D is a R.V.

h_D is a function of D and h_D is also a R.V.

What is Expectation of h_D ?

Expected classifier
 (Given A)

$$\bar{h} = E_{D \sim P^n} [h_D] = \int h_D P(D) dD$$

Expected test Error (Given A)

$$\Rightarrow E_{\substack{(x,y) \sim P \\ D \sim P^n}} [(h_D(x) - y)^2]$$

$$= \int \int \int (h_D(x) - y)^2 P(x, y) P(D) dx dy dD$$

Decomposition of Expected Test Error

$$\begin{aligned}
 E_{x,y,D} [[h_D(x) - y]^v] &= E_{x,y,D} [[(h_D(x) - \bar{h}(x)) + (\bar{h}(x) - y)]^v] \\
 &= E_{x,y,D} [(h_D(x) - \bar{h}(x))^v + 2 (h_D(x) - \bar{h}(x)) (\bar{h}(x) - y) \\
 &\quad + (\bar{h}(x) - y)^v] \\
 &= E_{x,y,D} [(h_D(x) - \bar{h}(x))^v] + 2 E_{x,y,D} [(h_D(x) - \bar{h}(x)) (\bar{h}(x) - y)] + E_{x,y,D} [(\bar{h}(x) - y)^v] \\
 &\stackrel{=}{=} E_{x,y,D} [(h_D(x) - \bar{h}(x)) (\bar{h}(x) - y)] = E_{x,y} [E_D [h_D(x) - \bar{h}(x)] (\bar{h}(x) - y)] \\
 &= E_{x,y} [(E_D [h_D(x)] - \bar{h}(x)) (\bar{h}(x) - y)] \\
 &\stackrel{=}{=} 0 \\
 &\stackrel{=}{=} E_{x,D} [(h_D(x) - \bar{h}(x))^v] + E_{x,y} [(\bar{h}(x) - y)^v] \\
 &\Rightarrow \underbrace{E_{x,D} [(h_D(x) - \bar{h}(x))^v]}_{\text{Variance}} \quad \textcircled{i}
 \end{aligned}$$

$$E_{x,y}[(\bar{h}(x) - y)^v] = E_{x,y}[(\bar{h}(x) - \bar{y}(x) + (\bar{y}(x) - y))^v]$$

$$= \underbrace{E_{x,y}[(\bar{h}(x) - \bar{y}(x))^v]}_{(2)} + \underbrace{E_x[(\bar{h}(x) - \bar{y}(x))^v]}_{(3)} + \underbrace{2 E_{x,y}[(\bar{h}(x) - \bar{y}(x))(\bar{y}(x) - y)]}_{\sim 0}$$

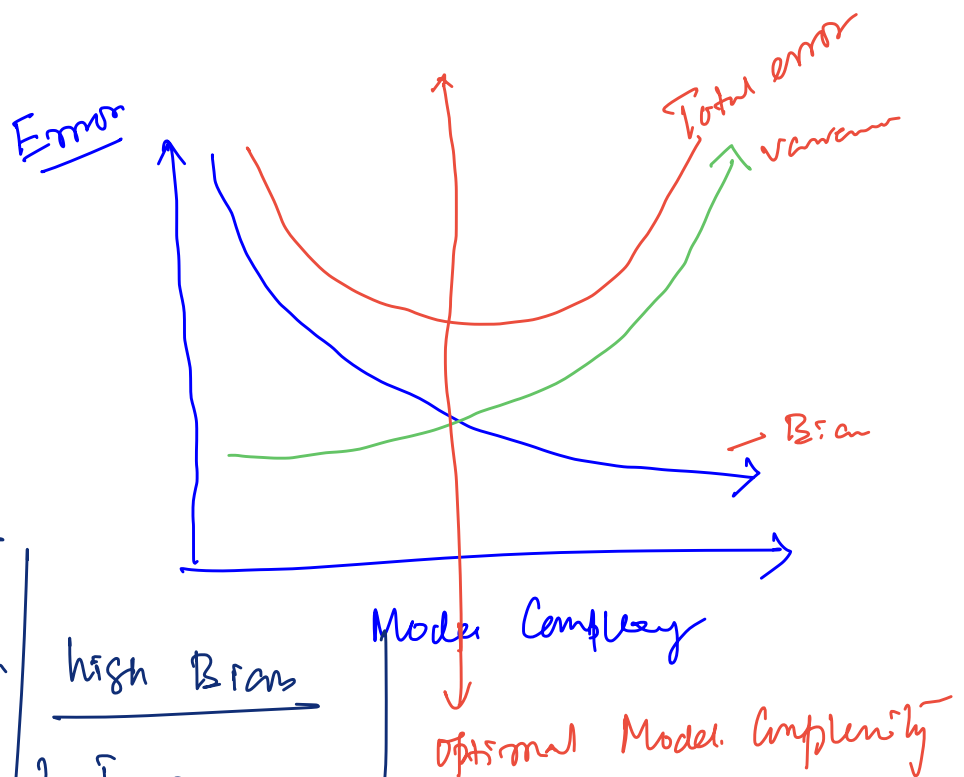
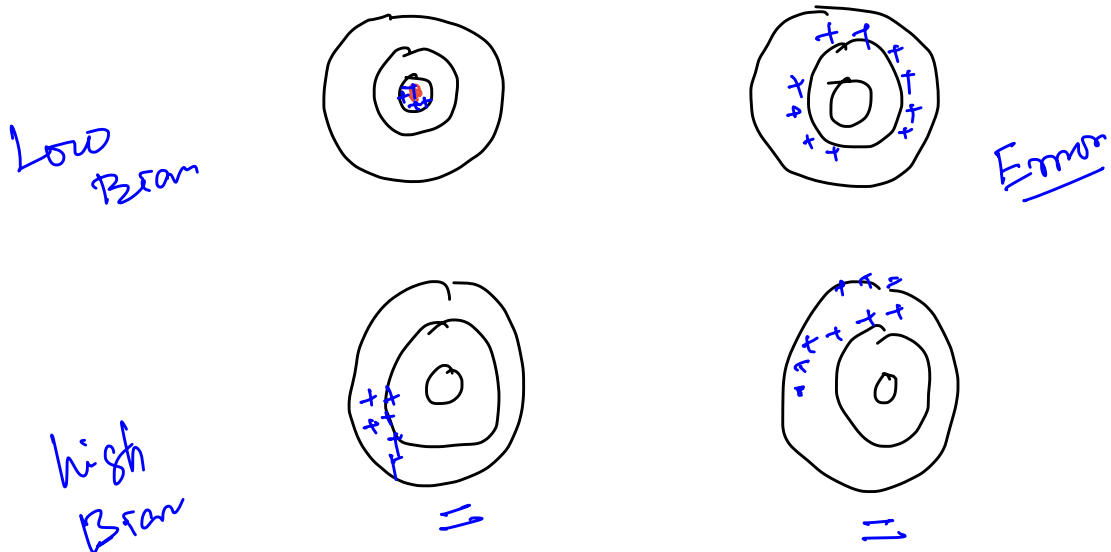
$$\begin{aligned} E_{x,y} [(\bar{h}(x) - \bar{y}(x))(\bar{y}(x) - y)] &= E_x [E_{y|x} [\bar{y}(x) - y] (\bar{h}(x) - \bar{y}(x))] \\ &= E_x [-E_{y|x} [y] + \bar{y}(x)] (\bar{h}(x) - \bar{y}(x)) \\ &= E_x [-\bar{y}(x) + \bar{y}(x)] (\bar{h}(x) - \bar{y}(x)) \\ &= 0 \end{aligned}$$

Expected Test Error:

$$= \underbrace{E_{x,D} [(h_0(x) - \bar{h}(x))^2]}_{\text{Variance}} + \underbrace{E_{x,y} [(\bar{y}(x) - y)^2]}_{\text{Noise}} + \underbrace{E_x [(\bar{h}(x) - \bar{y}(x))^2]}_{\text{Bias}^2}$$

Variance: Captures how much your classifier changes if you train on a different training set. How "Over-Specialized" is your classifier to a particular training set (overfitting)? If we have the best possible model for our training set, how far off we from the average classifier?

Bias: What is the inherent error that your model obtains from your classifier even with infinite training data? This is due to your classifier being "biased" to a particular kind of solution/Algo (e.g. linear classifier). Bias is inherent to your Algo.



High Variance

- Increase the size of the data
- Reduce model Complexity
- Bagging

high Bias

1. Increase model Complexity
2. Add more features.
3. Boosting